COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	Aug 26 2026	DATE SUBMITTED:	Aug 29 2026
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ACTIVITY NO. 5 **Consolidating Playbook plays**

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issue this command. Did something happen? Why?

```
kenzo@workstation:~/CPE212_Lim$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 410 bytes | 410.00 KiB/s, done.
From github.com:qknlim/CPE212_Lim
   374ca0d..dba0455  main       -> origin/main
Updating 374ca0d..dba0455
Fast-forward
 install_apache.yaml | 19 ++++++++++++++++++++
 1 file changed, 19 insertions(+)
 create mode 100644 install_apache.yaml
```

It produced an output of changing one file and 19 insertions as I did the previous activity in my personal computer. (I am doing this activity in Computer Laboratory)

Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

```
  GNU nano 6.2                                    install_apache.yaml
---
  - hosts: all
    become: true
    tasks:

    - name: Update Repository Index for Ubuntu
      apt:
        update_cache: yes
      when: ansible_distribution in ["Debian","Ubuntu"]

    - name: Install Apache2 Package
      apt:
        name: apache2
      when: ansible_distribution in ["Debian","Ubuntu"]

    - name: Add PHP support for apache
      apt:
        name: libapache2-mod-php
      when: ansible_distribution in ["Debian","Ubuntu"]
```

```
PLAY [all] ********************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************
ok: [192.168.56.7]
ok: [192.168.56.8]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This feature will be
 removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]

TASK [Update Repository Index for Ubuntu] *************************************************************************************
skipping: [192.168.56.9]
fatal: [192.168.56.8]: FAILED! => {"changed": false, "msg": "Failed to update apt cache: unknown reason"}
fatal: [192.168.56.7]: FAILED! => {"changed": false, "msg": "Failed to update apt cache: unknown reason"}

TASK [Install Apache2 Package] ************************************************************************************************
skipping: [192.168.56.9]

TASK [Add PHP support for apache] *********************************************************************************************
skipping: [192.168.56.9]

PLAY RECAP ********************************************************************************************************************
192.168.56.7               : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.8               : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.9               : ok=1    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

Edit the install_apache.yml file

kenzo@workstation: ~/CPE212_Lim

GNU nano 6.2 install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

  - name: Update Repository Index for Ubuntu
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: Install Apache2 Package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: Add PHP support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"

  - name: Update Repository Index for CentOS
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: Install Apache2 Package for CentOS
    dnf:
      name: httpd
    when: ansible_distribution == "CentOS"

  - name: Add PHP support for apache in CentOS
    dnf:
      name: php
    when: ansible_distribution == "CentOS"
```

```
PLAY [all] ********************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [192.168.56.8]
ok: [192.168.56.7]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]

TASK [Update Repository Index for Ubuntu] *************************************************************************************************
skipping: [192.168.56.9]
changed: [192.168.56.8]
changed: [192.168.56.7]

TASK [Install Apache2 Package] ************************************************************************************************************
skipping: [192.168.56.9]
changed: [192.168.56.8]
changed: [192.168.56.7]

TASK [Add PHP support for apache] *********************************************************************************************************
skipping: [192.168.56.9]
changed: [192.168.56.7]
changed: [192.168.56.8]

TASK [Update Repository Index for CentOS] *************************************************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

TASK [Install Apache2 Package for CentOS] *************************************************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

TASK [Add PHP support for apache in CentOS] ***********************************************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

PLAY RECAP ********************************************************************************************************************************
192.168.56.7               : ok=4    changed=3    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.8               : ok=4    changed=3    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.9               : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.

5.1 To activate, go to the CentOS VM terminal and enter the following:
systemctl status httpd

```
httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
  Drop-In: /usr/lib/systemd/system/httpd.service.d
           └─php-fpm.conf
   Active: inactive (dead)
     Docs: man:httpd.service(8)
```

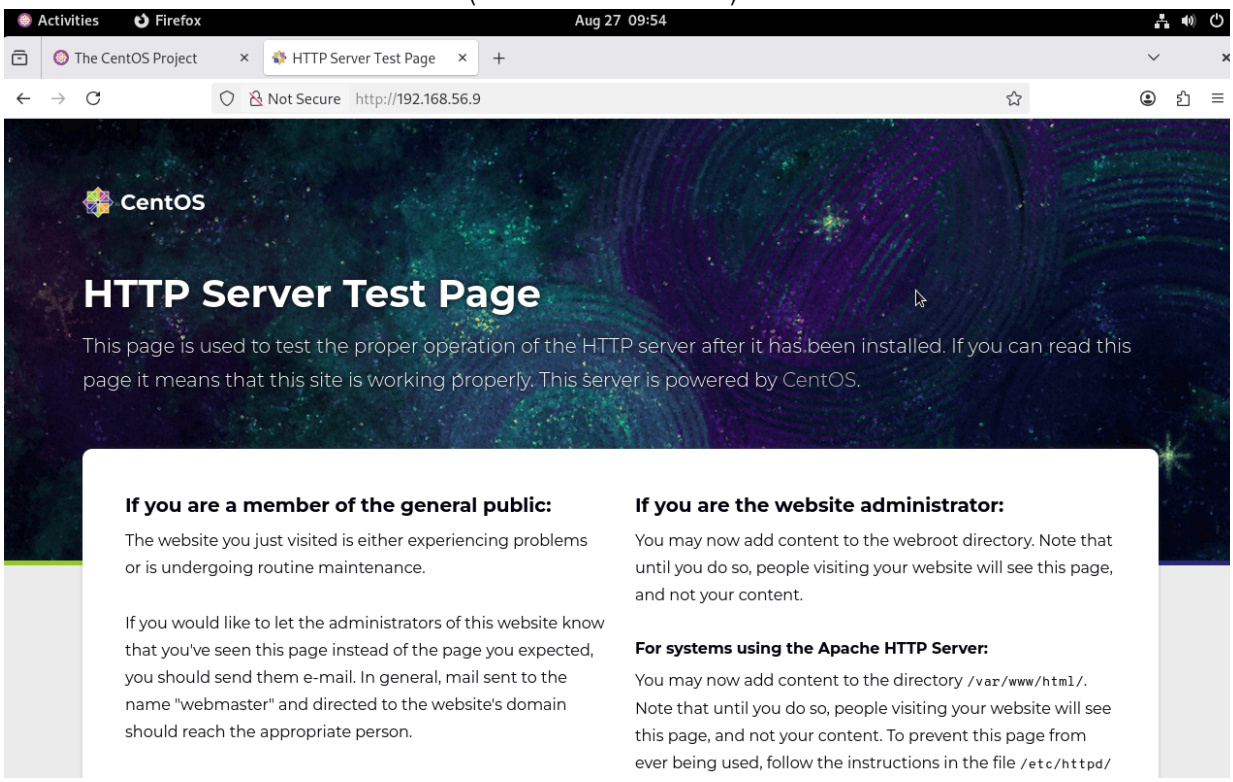
The result of this command tells you that the service is inactive.

5.2 Issue the following command to start the service:
sudo systemctl start httpd

(When prompted, enter the sudo password)
sudo firewall-cmd --add-port=80/tcp
(The result should be a success)

```
[kenzo@server3 ~]$ sudo firewall-cmd --add-port=80/tcp
[sudo] password for kenzo:
success
[kenzo@server3 ~]$ sudo systemctl start httpd
[kenzo@server3 ~]$
```

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)



Task 2: Refactoring playbook

Edit the playbook install_apache.yml.

```
 GNU nano 6.2                                                 install_apache.yaml
---
  - hosts: all
    become: true
    tasks:

    - name: Update Repository Index for Ubuntu
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: Install Apache2 Package and PHP support for apache in Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"


    - name: Update Repository Index for CentOS
      dnf:
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: Install Apache2 Package and PHP support for apache in CentOS
      dnf:
        name:
          - httpd
          - php
      when: ansible_distribution == "CentOS"
```

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook --ask-become-pass install_apache.yaml
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.7]
ok: [192.168.56.8]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward compatibility with
prior Ansible releases. A future Ansible release will default to using the discovered platform
 python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
 information. This feature will be removed in version 2.12. Deprecation warnings can be
disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]

TASK [Update Repository Index for Ubuntu] **************************************************
skipping: [192.168.56.9]
changed: [192.168.56.8]
changed: [192.168.56.7]

TASK [Install Apache2 Package and PHP support for apache in Ubuntu] ************************
skipping: [192.168.56.9]
ok: [192.168.56.8]
ok: [192.168.56.7]

TASK [Update Repository Index for CentOS] **************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

TASK [Install Apache2 Package and PHP support for apache in CentOS] ************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

PLAY RECAP *********************************************************************************
192.168.56.7               : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.8               : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

Edit the playbook install_apache.yml again.

```
  GNU nano 6.2                                          install_apache.yaml
---
  - hosts: all
    become: true
    tasks:

    - name: Install Apache2 Package and PHP support for apache in Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"


    - name: Install Apache2 Package and PHP support for apache in CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
      when: ansible_distribution == "CentOS"
```

```
kenzo@workstation:~/CPE212_Llm$ ansible-playbook --ask-become-pass install_apache.yaml
BECOME password:

PLAY [all] ***********************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************
ok: [192.168.56.7]
ok: [192.168.56.8]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]

TASK [Install Apache2 Package and PHP support for apache in Ubuntu] **************************************************************************
skipping: [192.168.56.9]
ok: [192.168.56.8]
ok: [192.168.56.7]

TASK [Install Apache2 Package and PHP support for apache in CentOS] **************************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

PLAY RECAP ***********************************************************************************************************************************
192.168.56.7               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.8               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

Edit the playbook install_apache.yml again

```
GNU nano 6.2                                                    install_ap
---
 - hosts: all
   become: true
   tasks:

   - name: Install Apache2 and PHP  Package for Ubuntu
     apt:
       name:
         - "{{ apache_package }}"
         - "{{ php_package }}"
       state: latest
       update_cache: yes
     when: ansible_distribution == "Ubuntu"

   - name: Install Apache2 Package for CentOS
     dnf:
       name:
         - httpd
         - php
       state: latest
       update_cache: yes
     when: ansible_distribution == "CentOS"
```

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook --ask-become-pass install_apache.yaml
BECOME password:

PLAY [all] *********************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.7]
ok: [192.168.56.8]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]

TASK [Install Apache2 Package and PHP support for apache in Ubuntu] ************************************************************
fatal: [192.168.56.8]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is
 undefined\n\nThe error appears to be in '/home/kenzo/CPE212_Lim/install_apache.yaml': line 6, column 7, but may\nbe elsewhere in the
 file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n    - name: Install Apache2 Package and PHP su
pport for apache in Ubuntu\n      ^ here\n"}
skipping: [192.168.56.9]
fatal: [192.168.56.7]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is
 undefined\n\nThe error appears to be in '/home/kenzo/CPE212_Lim/install_apache.yaml': line 6, column 7, but may\nbe elsewhere in the
 file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n    - name: Install Apache2 Package and PHP su
pport for apache in Ubuntu\n      ^ here\n"}

TASK [Install Apache2 Package and PHP support for apache in CentOS] ************************************************************
fatal: [192.168.56.9]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is
 undefined\n\nThe error appears to be in '/home/kenzo/CPE212_Lim/install_apache.yaml': line 16, column 7, but may\nbe elsewhere in th
e file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n    - name: Install Apache2 Package and PHP s
upport for apache in CentOS\n      ^ here\n"}

PLAY RECAP *********************************************************************************************************************
192.168.56.7               : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.8               : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.9               : ok=1    changed=0    unreachable=0    failed=1    skipped=1    rescued=0    ignored=0
```

Edit the inventory file (/etc/ansible/hosts)

kenzo@workstation: ~/CPE212_Lim

GNU nano 6.2 /etc/ansible/hosts

```
[ubuntu]
192.168.56.7 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.8 apache_package=apache2 php_package=libapache2-mod-php

[centos]
192.168.56.9 apache_package=httpd php_package=php
```

GNU nano 6.2 install_ap

```
---
- hosts: all
  become: true
  tasks:

  - name: Install Apache2 and PHP  Package for Ubuntu
    apt:
      name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: Install Apache2 Package for CentOS
    dnf:
      name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
      state: latest
      update_cache: yes
    when: ansible_distribution == "CentOS"
```

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook --ask-become-pass install_apache.yaml
BECOME password:

PLAY [all] *****************************************************************************************************************************

TASK [Gathering Facts] *****************************************************************************************************************
ok: [192.168.56.8]
ok: [192.168.56.7]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]

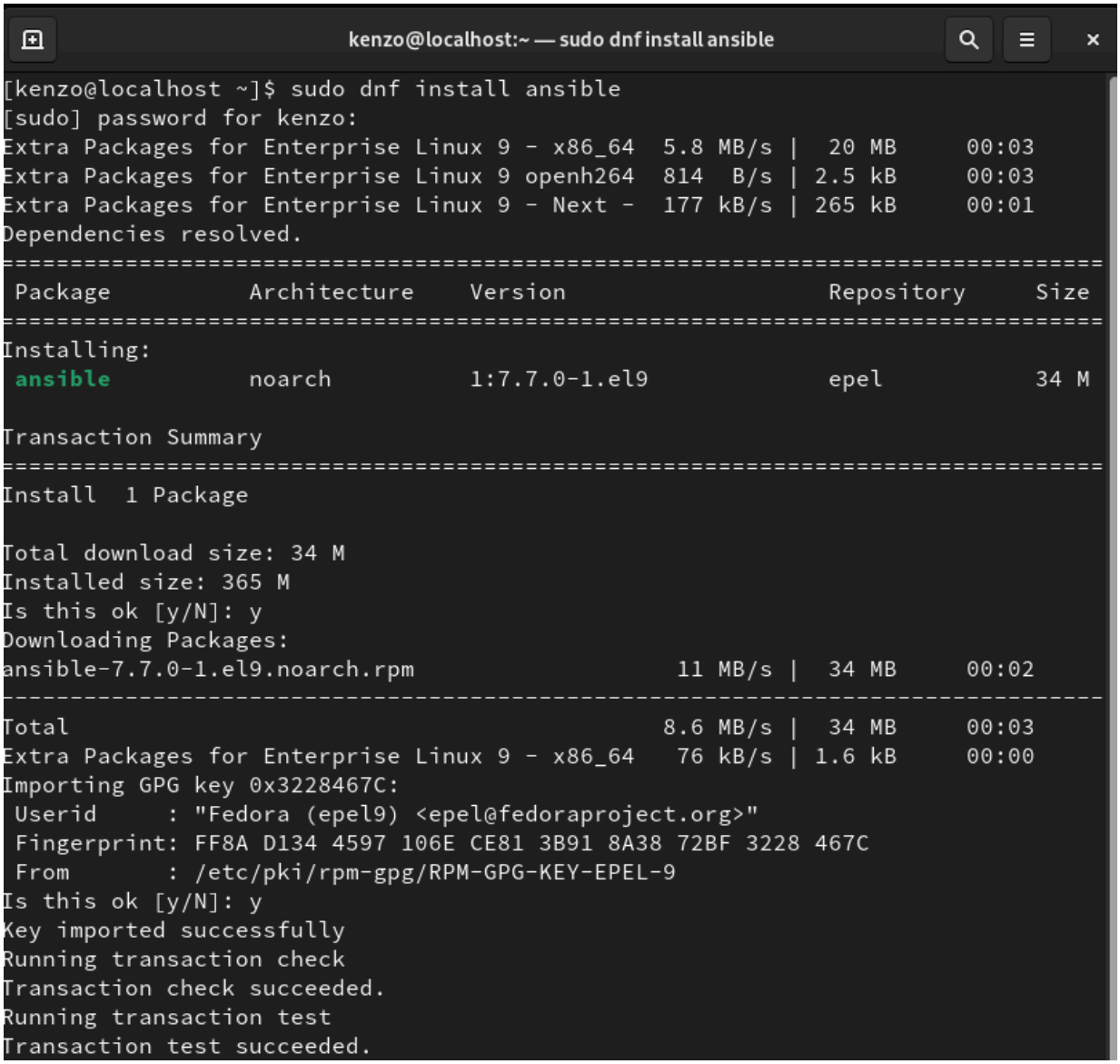
TASK [Install Apache2 Package and PHP support for apache in Ubuntu] ********************************************************************
skipping: [192.168.56.9]
ok: [192.168.56.7]
ok: [192.168.56.8]

TASK [Install Apache2 Package and PHP support for apache in CentOS] ********************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

PLAY RECAP *****************************************************************************************************************************
192.168.56.7               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.8               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a playbook that could do the previous tasks in Red Hat OS.
Installing Ansible in CentOS


kenzo@localhost:~ — sudo dnf install ansible
[kenzo@localhost ~]$ sudo dnf install ansible
[sudo] password for kenzo:
Extra Packages for Enterprise Linux 9 - x86_64 5.8 MB/s | 20 MB 00:03
Extra Packages for Enterprise Linux 9 openh264 814 B/s | 2.5 kB 00:03
Extra Packages for Enterprise Linux 9 - Next - 177 kB/s | 265 kB 00:01
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
ansible noarch 1:7.7.0-1.el9 epel 34 M
Transaction Summary
Install 1 Package
Total download size: 34 M
Installed size: 365 M
Is this ok [y/N]: y
Downloading Packages:
ansible-7.7.0-1.el9.noarch.rpm 11 MB/s | 34 MB 00:02
Total 8.6 MB/s | 34 MB 00:03
Extra Packages for Enterprise Linux 9 - x86_64 76 kB/s | 1.6 kB 00:00
Importing GPG key 0x3228467C:
Userid : "Fedora (epel9) <epel@fedoraproject.org>"
Fingerprint: FF8A D134 4597 106E CE81 3B91 8A38 72BF 3228 467C
From : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-9
Is this ok [y/N]: y
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.

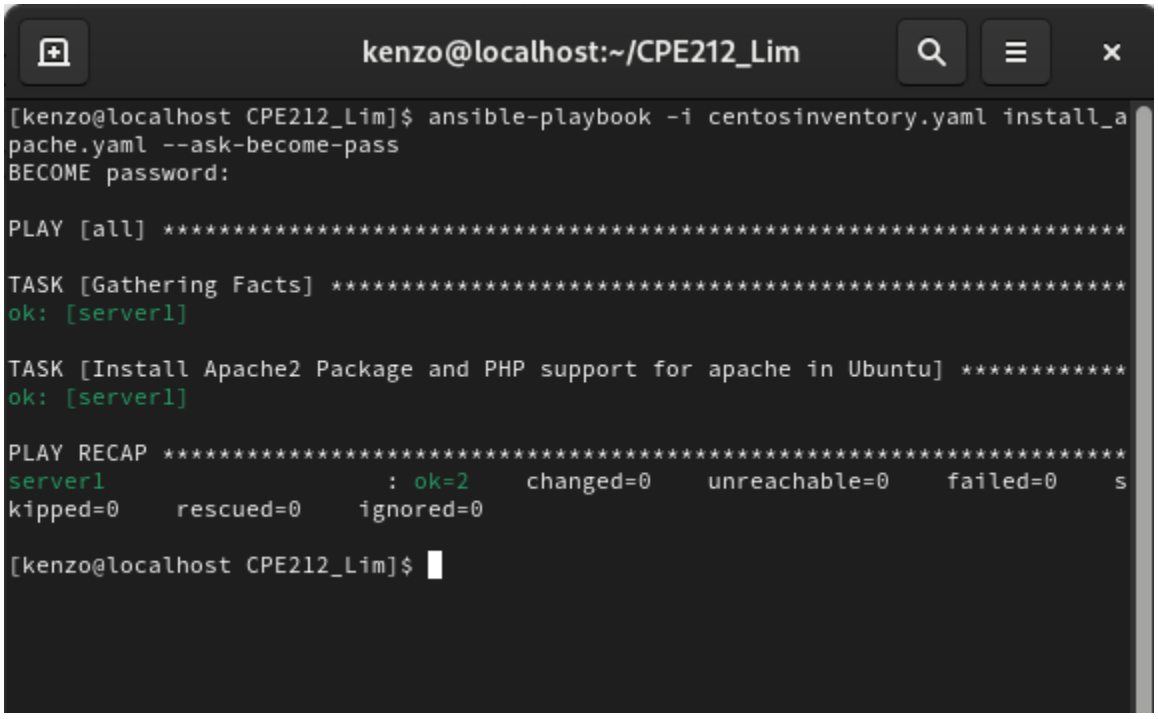
kenzo@localhost:~/CPE212_Lim — nano centosinventory.yaml

```
 GNU nano 5.6.1                    centosinventory.yaml
all:
  children:
    ubuntu:
      vars:
        ansible_user: kenzo
      hosts:
        server1:
          ansible_host: 192.168.56.105
          os_type: Ubuntu
          apache_package: apache2
          php_package: libapache2-mod-php
```

kenzo@localhost:~/CPE212_Lim — nano install_apache.yaml

```
 GNU nano 5.6.1                    install_apache.yaml
---
  - hosts: all
    become: true
    tasks:

    - name: Install Apache2 Package and PHP support for apache in Ubuntu
      apt:
        name:
          - "{{ apache_package}}"
          - "{{ php_package}}"
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"
```



I downloaded first git and ansible on CentOS using the commands
Sudo dnf install epel-release (sudo dnf install ansible later on) and sudo dnf install git
After that I utilized the previous playbook which is the install_apache.yaml to run it inside the CentOS control node. I realized that packages are more complex in CentOS as it has different names compared to the same package it has in Ubuntu.

1. **Why do you think refactoring of playbook codes is important?**

Refactoring codes in the playbook is important because it makes the playbook shorter and cleaner, as well as easy to understand. Removing repeated tasks using variables and generic modules like packages. The same configuration also works on different OS like CentOS and Ubuntu.

2. **When do we use the "when" command in playbook?**

We use the when command when we want the task to run only if a specific condition is true. An example would be when we have two different operating systems, we can use when: ansible_distribution == "Ubuntu" to make the task run only on the Ubuntu machines. This allows the playbook to handle different operating systems.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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